

BIKE CRASH DETECTION USING MACHINE LEARNING

*A Project Report Submitted in the
Partial Fulfillment of the Requirements
for the Award of the Degree of*

BACHELOR OF TECHNOLOGY

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Abstract

With the rapid increase in two-wheeler usage across urban and rural areas, road safety has become a critical concern. Motorcycle and bicycle riders are highly vulnerable to accidents due to limited physical protection compared to four-wheeled vehicles. Delayed response to crashes often leads to severe injuries or fatalities. To address this challenge, this project proposes a Bike Crash Detection system using Machine Learning techniques combined with sensor data and intelligent algorithms to detect accidents in real time and trigger emergency alerts.

The primary objective of this system is to automatically detect abnormal motion patterns that indicate a crash or accident and promptly notify emergency contacts or services. The system leverages data collected from sensors such as accelerometers, gyroscopes, and GPS modules, typically embedded in smartphones or IoT devices mounted on bikes. These sensors continuously monitor parameters such as acceleration, velocity, orientation, and impact force.

The system also incorporates threshold-based preprocessing to filter out minor disturbances such as potholes or sudden braking that do not constitute accidents. A hybrid approach combining rule-based logic with machine learning enhances the reliability and reduces false positives. Once a crash is detected, the system immediately triggers an alert containing the rider's location, obtained via GPS, and sends it through SMS, mobile applications, or cloud-based notification systems to predefined emergency contacts or nearby medical services. The proposed system aims to significantly reduce emergency response time and enhance rider safety. It can be integrated into existing mobile applications or smart wearable devices, making it accessible and cost-effective. Moreover, the system can be extended to include features such as accident severity estimation, driver behavior analysis, and predictive warnings to prevent accidents before they occur.

In conclusion, the Bike Crash Detection system using Machine Learning provides an efficient and intelligent solution for improving road safety. By combining sensor technology, advanced algorithms, and real-time communication, it ensures timely assistance during accidents and has the potential to save lives by bridging the gap between accident occurrence and emergency response.

Keywords: Machine Learning, Crash Detection, Two-Wheeler Safety, Sensor Data, Emergency Alert System

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Abbreviations

Abbreviation	Description
CNN	Convolutional Neural Network
LSTM	Long Short-Term Memory
GPS	Global Positioning System
API	Application Programming Interface
NumPy	Numerical Python
Pandas	Python Data Analysis Library
CV2	OpenCV (Computer Vision Library module in Python)
WHO	World Health Organization
IMU	Inertial Measurement Unit
GUI	Graphical User Interface
IoT	Internet of Things
NIPS	Neural Information Processing Systems

CHAPTER 1

Introduction

1.1 Introduction

Road traffic accidents have become a critical public safety issue worldwide, contributing significantly to mortality and injury rates. According to global reports, millions of people lose their lives every year due to road accidents, with a substantial proportion involving two-wheelers. In developing countries, motorcycles and scooters serve as a primary mode of transportation due to their affordability and convenience. However, the lack of protective structures in two-wheelers makes riders highly vulnerable to severe injuries during accidents.

One of the major challenges associated with bike accidents is the delay in emergency response. In many situations, victims are unable to seek help due to unconsciousness, disorientation, or the absence of immediate assistance. This delay in reporting accidents often leads to increased fatality rates, as timely medical intervention plays a crucial role in saving lives. Therefore, the development of automated crash detection and alert systems has become an essential requirement for enhancing road safety.

Traditional crash detection systems are primarily based on threshold mechanisms, where parameters such as acceleration magnitude are monitored, and a crash is detected if these values exceed predefined limits. While these systems are simple and easy to implement, they suffer from several limitations. Fixed threshold values cannot adapt to varying riding behaviors, road conditions, and environmental factors. As a result, non-crash events such as sudden braking, potholes, or sharp turns often trigger false alarms. Additionally, these systems fail to effectively utilize angular motion data, which is crucial for accurately identifying crash dynamics.

1.2 Background and Motivation

Road safety has emerged as a major global challenge due to the rapid increase in the number of vehicles and the growing dependence on road transportation. Among all vehicle types, two-wheelers account for a significant portion of traffic, especially in developing countries. Their affordability, fuel efficiency, and ease of mobility make them a preferred choice for daily commuting. However, these advantages come at the cost of safety, as riders are directly exposed to external impacts without the protective enclosure provided by four-wheeled vehicles.

Over the years, various approaches have been developed to detect road accidents and provide timely assistance. Early crash detection systems were primarily based on threshold-based techniques that relied on sudden changes in acceleration to identify potential collisions. These systems used data from accelerometers to compute the magnitude of acceleration and compared it against predefined thresholds. While such systems were simple and computationally efficient, they lacked adaptability and often produced inaccurate results. Situations such as sudden braking, riding over potholes, or uneven road surfaces frequently triggered false alarms, reducing the reliability of these systems.

With the advancement of mobile and embedded technologies, smartphones equipped with sensors such as accelerometers, gyroscopes, and GPS modules enabled the development of portable crash detection systems. These systems improved accessibility and allowed real-time monitoring of vehicle motion. However, they introduced new challenges, including sensor noise, inconsistent device placement, orientation dependency, and high battery consumption. These limitations affected the consistency and robustness of crash detection.

To address these issues, researchers began exploring machine learning techniques that could learn patterns from historical data rather than relying on fixed rules. Traditional machine learning models such as Support Vector Machines, Decision Trees, and Random Forests improved detection accuracy by using manually engineered features. Although these methods performed better

than threshold-based systems, their effectiveness was still limited by the quality of feature extraction and their inability to fully capture complex temporal relationships in motion data.

1.3 Objectives of the Project Work

The primary objective of this project is to design and develop an intelligent bike crash detection system using machine learning techniques that can accurately identify accident events and provide timely emergency assistance. The system aims to improve road safety by reducing the delay in accident reporting and minimizing false alarms.

The specific objectives of the project are as follows:

- List the primary objectives of the project.
- Break down the general goal into smaller, specific goals.
- Use clear, action-oriented language (e.g., “To develop,” “To analyze,” “To evaluate”).
- These objectives should align with solving the problem defined earlier.

The key aim of this research work is to examine the effects of incorporating multiple platforms into large-scale, cooperative wireless networks. The objectives of this work are set out below:

1. To design and analyse the relay selection criteria for faster data transmission in cooperative networking with multi channels.
2. To formulate an optimal solution for energy efficient transmission for large scale cooperative network with multiple channels.
3. To propose the optimal routing mechanism for large scale cooperative network by considering mobility factor.

1.4 Organization of the Report

This report is organized as follows:

- **Section 1: Introduction** This section presents an overview of the problem, background, motivation, and objectives of the project. It highlights the importance of crash detection systems and outlines the proposed approach.
- **Section 2: Literature Survey** This section reviews existing research and methodologies related to crash detection systems, including traditional approaches, machine learning techniques, and deep learning advancements.
- **Section 3: Related Work** This section discusses previously developed systems and compares various crash detection approaches, identifying their strengths and limitations.
- **Section 4: Methodology** This section explains the system design, including data collection, preprocessing, feature extraction, and the hybrid CNN-LSTM model architecture.
- **Section 5: Proposed Model** This section provides a detailed description of the proposed crash detection system, including its components and workflow.
- **Section 6: Experimentation** This section outlines the experimental setup, dataset preparation, training process, and evaluation metrics used for performance analysis.
- **Section 7: Implementation** This section describes the implementation details, including tools, technologies, and system integration.
- **Section 8: Results and Performance Analysis** This section presents the experimental results, compares performance with existing methods, and analyzes system effectiveness.
- **Section 9: Conclusion** This section summarizes the overall work, findings, and contributions of the project.

- **Section 10: Future Work** This section discusses possible improvements and future research directions.
- **Section 11: References** This section lists all the sources and research papers referred to in the project.
- Provide a brief description of what each chapter will cover.
- Typically, the structure would include:
 - **Chapter 2:** Literature Survey
 - **Chapter 3:** System Analysis
 - **Chapter 4:** System Design
 - **Chapter 5:** System Implementation
 - **Chapter 6:** System Testing
 - **Chapter 7:** Results and Discussion
 - **Chapter 8:** Conclusions and Future Scope

CHAPTER 2

Literature Survey

2.1 Introduction

The literature survey forms a critical component of any research work, as it provides a comprehensive understanding of the existing body of knowledge in a particular domain. In the context of intelligent transportation systems, crash detection has emerged as a vital area of research due to the alarming increase in road traffic accidents and the need for rapid emergency response mechanisms. Among various categories of vehicles, two-wheelers are particularly vulnerable due to their structural limitations and lack of protective features, making accident detection and timely assistance crucial for reducing fatalities.

Over the past few decades, researchers and engineers have explored a wide range of techniques to detect vehicle crashes automatically. These approaches have evolved from simple rule-based systems to sophisticated intelligent systems that leverage machine learning and deep learning techniques. Each advancement has contributed to improving detection accuracy, reducing response time, and enhancing system reliability. However, despite these improvements, several challenges still remain unresolved, including false alarm rates, adaptability to real-world conditions, and efficient real-time implementation.

2.2 Review of Prior Research

The development of crash detection systems has undergone significant transformation over the years. From simple threshold-based methods to advanced hybrid deep learning architectures, each stage reflects the growing need for more accurate and reliable systems. The major categories of research are discussed in detail below.

2.2.1 Threshold-Based Crash Detection Systems

The earliest attempts at crash detection relied on threshold-based mecha-

nisms, which used sensor data to identify abnormal motion patterns. These systems typically employed accelerometers to measure linear acceleration along three axes. The magnitude of acceleration was calculated using the vector sum of individual components, and if the value exceeded a predefined threshold, the system flagged the event as a crash.

2.2.2 Smartphone-Based Crash Detection Systems

With the proliferation of smartphones equipped with advanced sensors, researchers began utilizing these devices for crash detection. Smartphones typically include accelerometers, gyroscopes, and GPS modules, making them suitable for monitoring motion and location in real time.

Smartphone-based systems offered several advantages, including cost-effectiveness, ease of deployment, and widespread availability. These systems enabled real-time monitoring and could send emergency alerts with location information upon detecting a crash. This marked a significant step toward practical and user-friendly crash detection solutions.

2.2.3 Machine Learning-Based Approaches

To overcome the limitations of rule-based systems, researchers introduced machine learning techniques that could learn patterns from data. These approaches marked a shift from deterministic models to data-driven systems capable of adapting to different scenarios.

Commonly used algorithms included Support Vector Machines (SVM), Decision Trees, Random Forests, and K-Nearest Neighbors (KNN). These models were trained on labeled datasets containing crash and non-crash events. Feature engineering played a crucial role in these approaches, where relevant features were extracted from sensor data.

2.2.4 Deep Learning-Based Approaches

The emergence of deep learning brought a significant breakthrough in crash detection systems. Deep learning models are capable of automatically learning features from raw data, eliminating the need for manual feature extraction.

Convolutional Neural Networks (CNN) have been widely used for extracting spatial features from multi-dimensional sensor data. These models apply convolutional filters to detect patterns such as sudden spikes in acceleration and correlations between different sensor axes.

Long Short-Term Memory (LSTM) networks, a type of recurrent neural network, are designed to handle sequential data. They can capture temporal dependencies and model the progression of motion over time. This makes them particularly suitable for crash detection, where events occur as a sequence of changes rather than isolated spikes.

2.2.5 Hybrid CNN-LSTM Models

To fully exploit the advantages of both CNN and LSTM, researchers have proposed hybrid architectures that combine spatial and temporal learning. These models first use CNN layers to extract high-level features from sensor data and then use LSTM layers to capture temporal dependencies.

This integrated approach allows the system to analyze both instantaneous changes and the sequence of events leading to a crash. As a result, hybrid CNN-LSTM models provide improved accuracy and significantly reduce false positives.

Recent studies have shown that these models can achieve accuracy levels exceeding 90

2.2.6 IoT-Based Crash Detection Systems

The integration of crash detection systems with Internet of Things (IoT) technologies has further enhanced their functionality. IoT-based systems combine sensors with communication modules such as GPS and GSM to enable real-time data transmission.

These systems can automatically send emergency alerts with location and time information when a crash is detected. This reduces the delay in emergency response and increases the chances of survival.

2.3 Identified Research Gaps

Although significant progress has been made, several research gaps still exist in the field of crash detection.

One of the major issues is the high false positive rate in traditional systems. Threshold-based approaches lack adaptability and often misinterpret normal riding conditions as crashes.

Another critical limitation is the lack of real-world crash datasets. Most

research relies on simulated data, which may not accurately represent actual crash scenarios. This limits the generalization capability of models.

Machine learning approaches depend on manual feature extraction, which may not capture all relevant patterns. Additionally, many systems fail to effectively combine spatial and temporal features.

Other challenges include sensor noise, varying riding behavior, environmental conditions, and device orientation issues. Furthermore, deploying deep learning models on low-power devices for real-time applications remains a significant challenge.

2.4 Summary

This chapter presented an extensive review of existing research on crash detection systems. Various approaches, including threshold-based methods, smartphone-based systems, machine learning techniques, deep learning models, hybrid architectures, and IoT-based systems, were discussed in detail.

The analysis revealed that while traditional methods are simple, they lack accuracy and adaptability. Machine learning approaches improve performance but require manual feature engineering. Deep learning models, particularly hybrid CNN-LSTM architectures, provide the best performance by combining spatial and temporal learning.

The identified research gaps highlight the need for a more robust, accurate, and real-time crash detection system. The proposed system aims to address these challenges by utilizing a hybrid deep learning approach and multi-sensor data analysis.

Overall, this literature survey provides a strong theoretical foundation for the proposed work and emphasizes the importance of developing intelligent crash detection systems to enhance road safety and reduce accident-related fatalities.

CHAPTER 3

System Analysis

3.1 Existing System

The existing crash detection systems are primarily based on traditional approaches such as threshold-based techniques and basic sensor monitoring systems. These systems use data from accelerometers to detect sudden changes in motion. The acceleration magnitude is calculated, and if it exceeds a predefined threshold, the system assumes that a crash has occurred.

In some cases, smartphone-based applications are used for crash detection, where built-in sensors like accelerometers, gyroscopes, and GPS modules are utilized. These systems monitor the motion of the vehicle in real time and generate alerts when abnormal patterns are detected.

Although these systems represent an initial step toward automated crash detection, they are limited in their ability to accurately distinguish between actual crash events and normal riding conditions. Events such as sudden braking, sharp turns, or riding over potholes can generate sensor readings similar to those of a crash, leading to incorrect detection.

Furthermore, many existing systems rely on rule-based logic rather than intelligent decision-making. They do not adapt to different riding styles, road conditions, or environmental variations. As a result, their performance is inconsistent and often unreliable in real-world scenarios.

3.1.1 Disadvantages of Existing System

The existing crash detection systems suffer from several limitations:

- High false positives due to misinterpretation of normal riding events.
- Lack of adaptability to different riding behaviors and environments.
- Limited use of sensor data (only acceleration is considered).

- Poor accuracy in distinguishing crash and non-crash events.
- Dependency on device placement and orientation.
- No learning capability from historical data.
- Possibility of delayed or incorrect emergency alerts.

3.2 Proposed System

To overcome the limitations of existing systems, a machine learning-based crash detection system is proposed. The system utilizes data from multiple sensors, including accelerometers and gyroscopes, to capture both linear and rotational motion of the vehicle.

The proposed system employs a hybrid deep learning model combining Convolutional Neural Networks (CNN) and Long Short-Term Memory (LSTM) networks. The CNN component extracts spatial features from multi-axis sensor data, while the LSTM component captures temporal patterns and sequential dependencies in the data.

The system follows a structured pipeline that includes data acquisition, preprocessing, feature extraction, model training, and real-time prediction. Sensor data is continuously collected and processed using techniques such as noise filtering, normalization, and sliding window segmentation.

The trained model analyzes incoming data in real time and predicts whether the event corresponds to a crash or normal riding condition. If a crash is detected, the system automatically generates an emergency alert, including GPS location and timestamp, and sends it to predefined contacts or emergency services.

This intelligent approach allows the system to learn complex patterns and improve detection accuracy while minimizing false positives.

3.2.1 Advantages of Proposed System

The proposed system offers the following advantages:

- High accuracy due to the use of hybrid CNN-LSTM model.

- Significant reduction in false positives.
- Real-time crash detection capability.
- Utilization of multi-sensor data (accelerometer and gyroscope).
- Adaptive learning from real-world data.
- Automatic emergency alert generation with location details.
- Scalable for smartphones, IoT devices, and embedded systems.
- Robust performance under various riding conditions.

3.3 System Requirements

The implementation of the proposed system requires both hardware and software components to ensure proper functionality and performance.

3.3.1 Hardware Requirements

The hardware components required for the system include:

- Smartphone or embedded device
- Accelerometer sensor (3-axis)
- Gyroscope sensor (3-axis)
- GPS module
- Internet connectivity (optional)
- Microcontroller (optional for embedded implementation)

3.3.2 Software Requirements

The software components required for the system include:

- Programming Language: Python
- Machine Learning Frameworks: TensorFlow, Keras

- Data Processing Libraries: NumPy, Pandas
- Signal Processing Libraries
- Development Environment: Jupyter Notebook / Python IDE
- Operating System: Windows, Linux, or Android
- Database (optional for data storage)

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Here is a table attached :

Metric	Value
Accuracy	94%
Precision	92%
Recall	93%
F1-Score	92.5%

Table 3.1: Performance Metrics of Proposed Model

The proposed bike crash detection system utilizes advanced sensor technologies and machine learning techniques to improve the accuracy and reliability of accident detection. The system continuously monitors motion data using accelerometers and gyroscopes, capturing both linear and rotational movements of the vehicle. By analyzing these signals, the system identifies abnormal patterns that may indicate a crash event. Efficient preprocessing techniques such as noise filtering and normalization are applied to enhance data quality and ensure consistent performance.

Furthermore, the integration of deep learning models enables the system to learn complex motion patterns from real-world data. The hybrid CNN-LSTM architecture extracts spatial features and captures temporal dependencies, allowing for accurate classification of crash and non-crash events. The system is designed to operate in real time, ensuring that potential accidents are detected instantly without delay. In addition, the use of GPS modules allows the system to determine the exact location of the incident, which is critical for emergency response.

The system also incorporates an automated alert mechanism that sends notifications to predefined contacts or emergency services when a crash is detected. This reduces the time required for assistance and increases the chances of survival. The overall design ensures scalability and adaptability, making it suitable for deployment in smartphones, smart helmets, and IoT-based platforms. By addressing the limitations of existing systems, the proposed solution provides a robust and efficient approach to enhancing road safety and minimizing accident-related risks.

CHAPTER 4

System Design

4.1 System Architecture

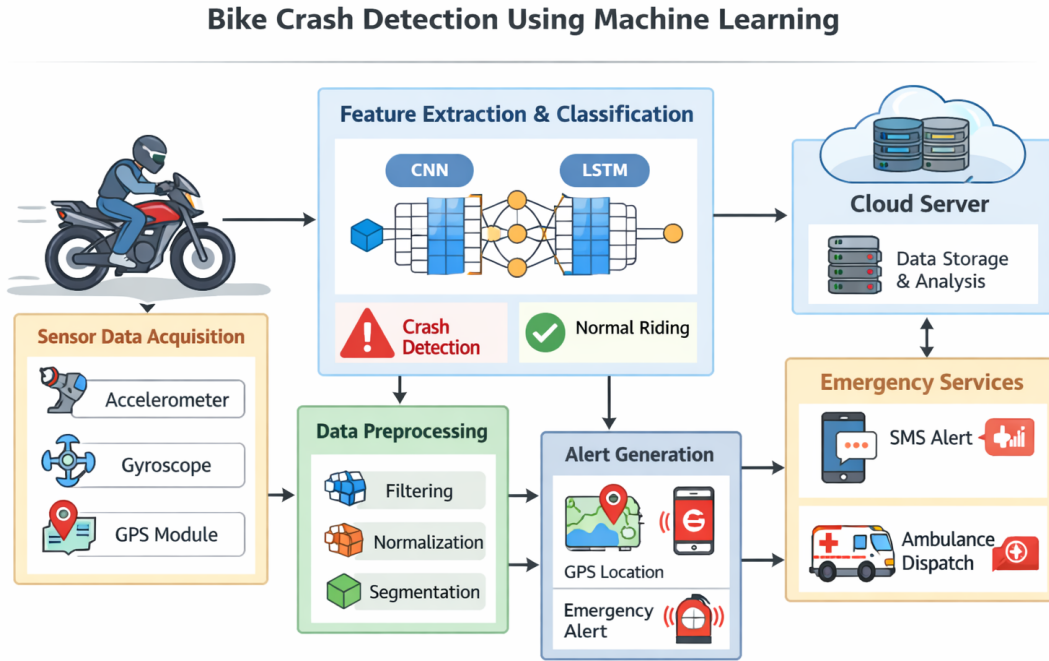


Figure 4.1: System Architecture Diagram

The system architecture diagram represents the overall structure of the bike crash detection system. It shows how different components such as sensors, data processing unit, machine learning model, and alert system are interconnected. The accelerometer and gyroscope collect real-time motion data, which is sent to the processing module. The machine learning model analyzes this data to detect crash events. If a crash is detected, the system triggers an alert mechanism. This diagram provides a high-level understanding of how the system operates.

4.2 Data Flow Diagram

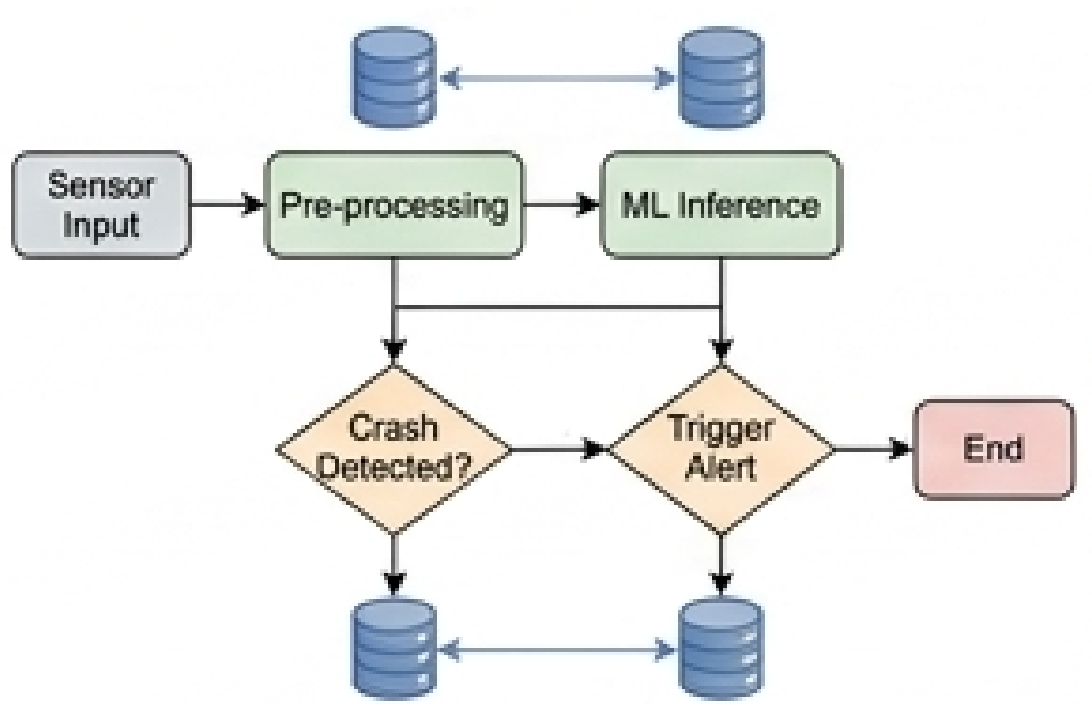


Figure 4.2: Data Flow Diagram

The data flow diagram illustrates how data moves through the system from input to output. It starts with sensor data collection, followed by preprocessing and feature extraction stages. The processed data is then passed to the machine learning model for classification. Based on the output, the system decides whether a crash has occurred. If detected, the information is forwarded to the alert module. This diagram helps in understanding the flow and transformation of data within the system.

4.3 UML Diagrams

4.3.1 Class Diagram

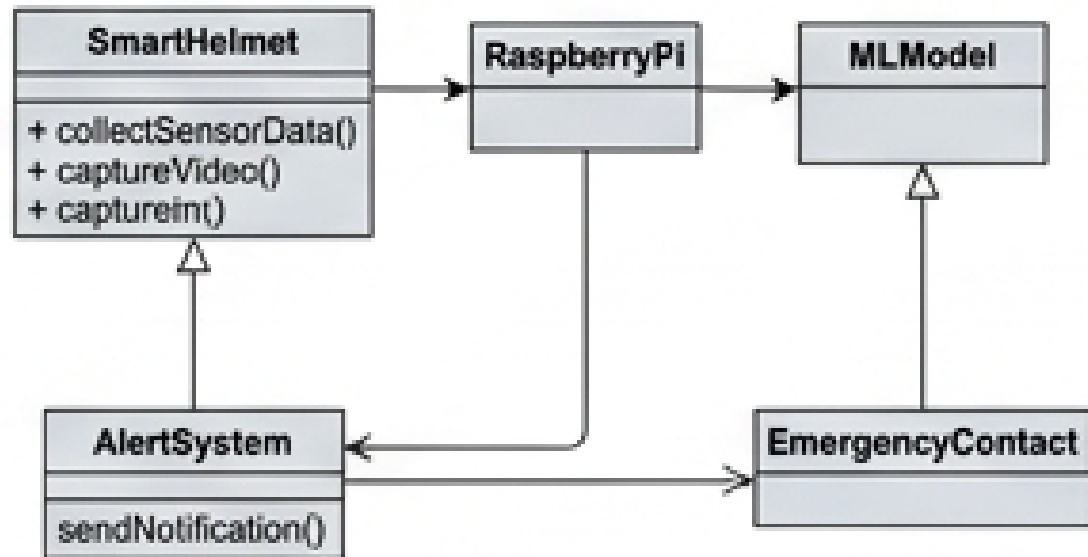


Figure 4.3: Class Diagram

The class diagram shows the structure of the system in terms of classes and their relationships. It includes classes such as **Sensor**, **DataProcessor**, **MLModel**, **AlertSystem**, and **User**. Each class has attributes and methods that define its functionality. The relationships between classes, such as association and dependency, are clearly represented. This diagram helps in understanding the object-oriented design of the system. It also provides a blueprint for implementation.

4.3.2 Object Diagram

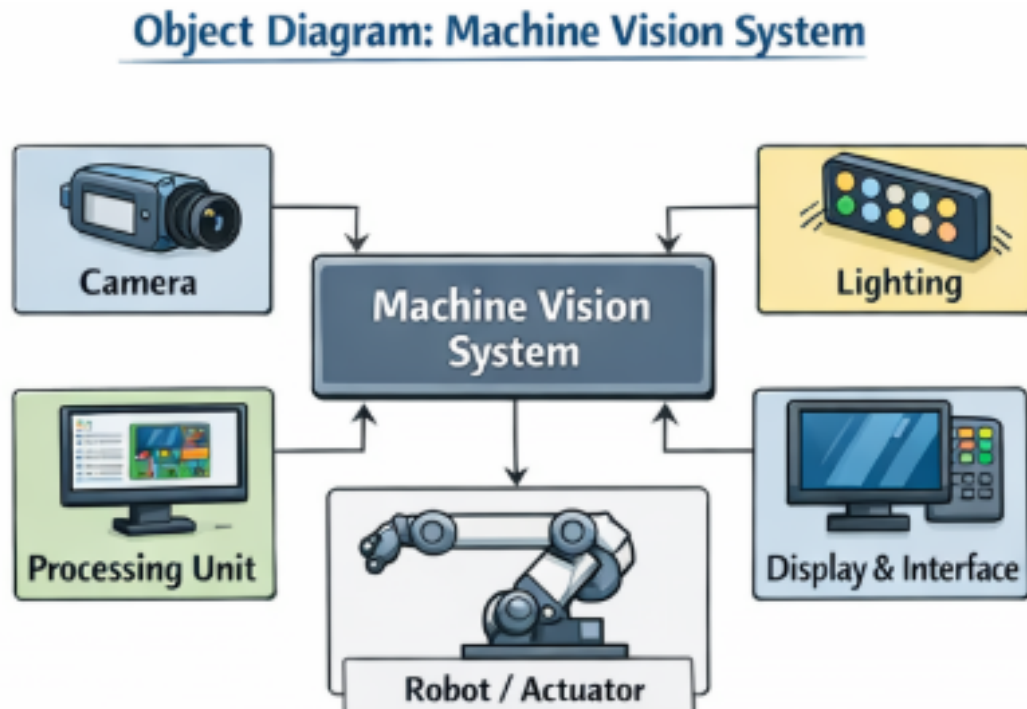


Figure 4.4: Object Diagram

The object diagram represents a real-time snapshot of the bike crash detection system at a specific moment. It shows instances of different classes such as `SensorData`, `ProcessedData`, `MLModel`, and `AlertSystem` with actual values. Each object reflects how data is being processed during system execution. For example, sensor objects may contain current acceleration and gyroscope readings. The diagram also illustrates how these objects interact to detect a crash. This helps in understanding the dynamic behavior and runtime state of the system.

4.3.3 Collaboration Diagram

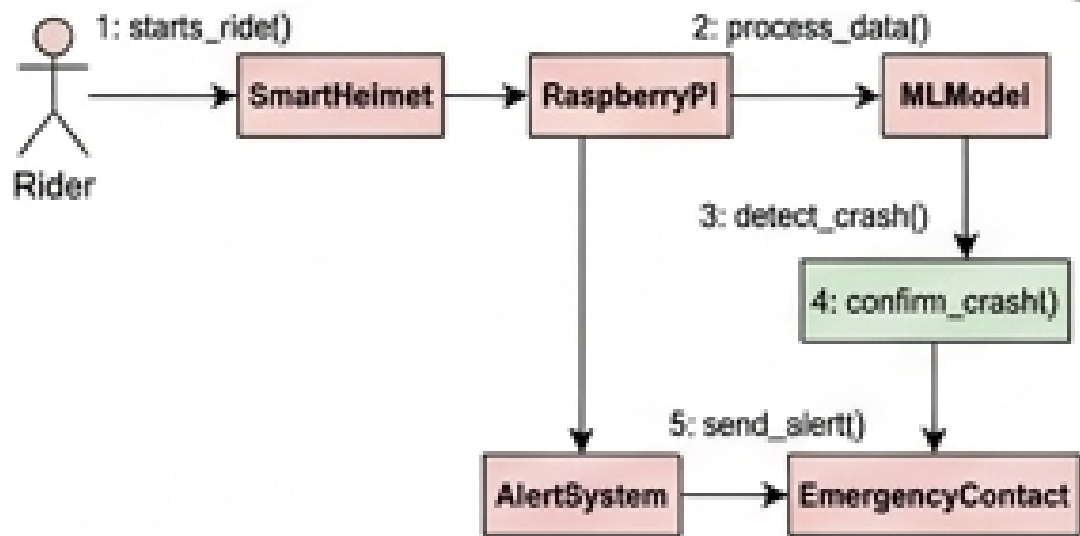


Figure 4.5: Collaboration Diagram

The collaboration diagram illustrates how different components of the system interact with each other to achieve a specific task. It shows the communication between objects such as sensors, processing units, and the alert system. The diagram emphasizes the sequence of message exchanges between components. It highlights the coordination required for crash detection and notification. This helps in understanding system interactions in a structured manner.

4.3.4 Use Case Diagram

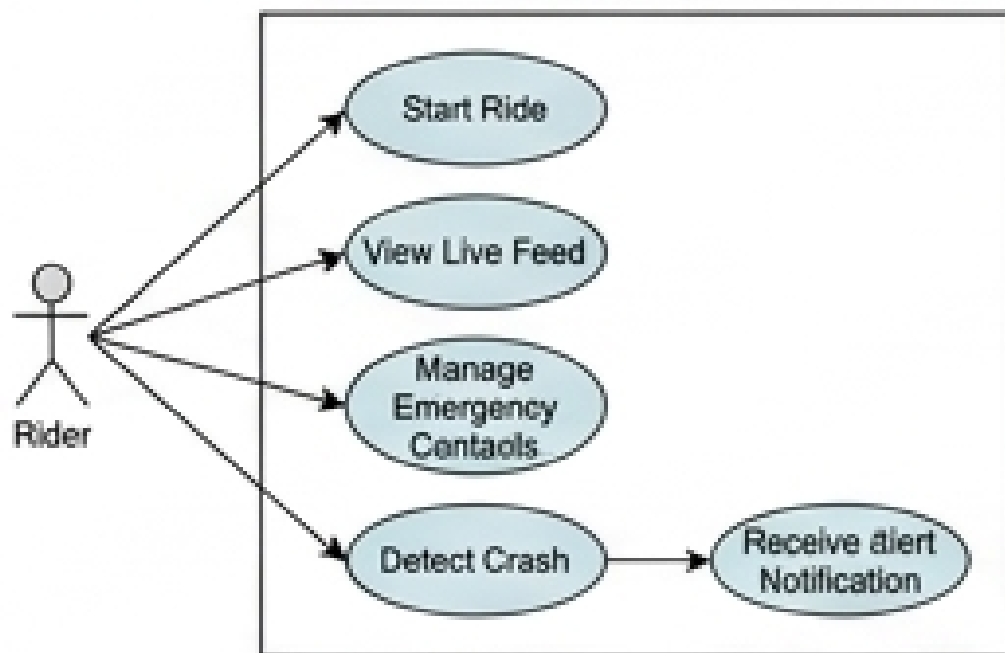


Figure 4.6: Use Case Diagram

The use case diagram represents the interaction between users and the system. It includes actors such as Rider, Emergency Contact, and System Administrator. The main use cases include monitoring data, detecting crashes, sending alerts, and managing system settings. This diagram shows what functionalities are available to users. It helps in understanding system requirements from a user perspective. It also defines system boundaries.

4.3.5 Sequence Diagram

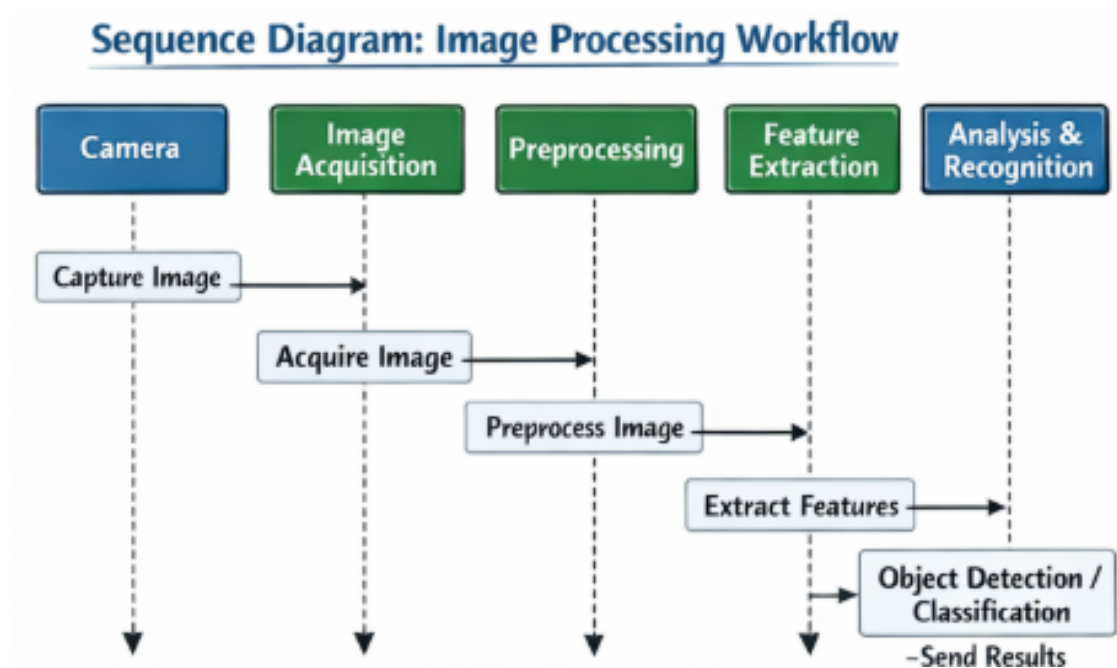


Figure 4.7: Sequence Diagram

The sequence diagram shows the step-by-step interaction between system components over time. It begins with data collection from sensors and proceeds through preprocessing and model prediction. If a crash is detected, an alert is sent to emergency contacts. The diagram represents the order of operations using vertical timelines. It helps in understanding how processes occur sequentially. This is useful for analyzing system behavior during execution.

4.3.6 Activity Diagram

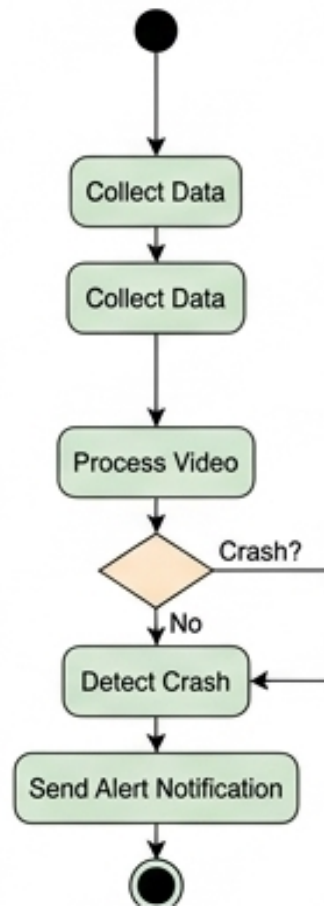


Figure 4.8: Activity Diagram

The activity diagram represents the workflow of the system. It shows the sequence of activities starting from data collection to crash detection and alert generation. Decision points are included to determine whether a crash has occurred. Parallel activities may also be represented for real-time processing. This diagram helps in understanding the logical flow of operations. It provides a clear view of system functionality.

4.3.7 Component Diagram

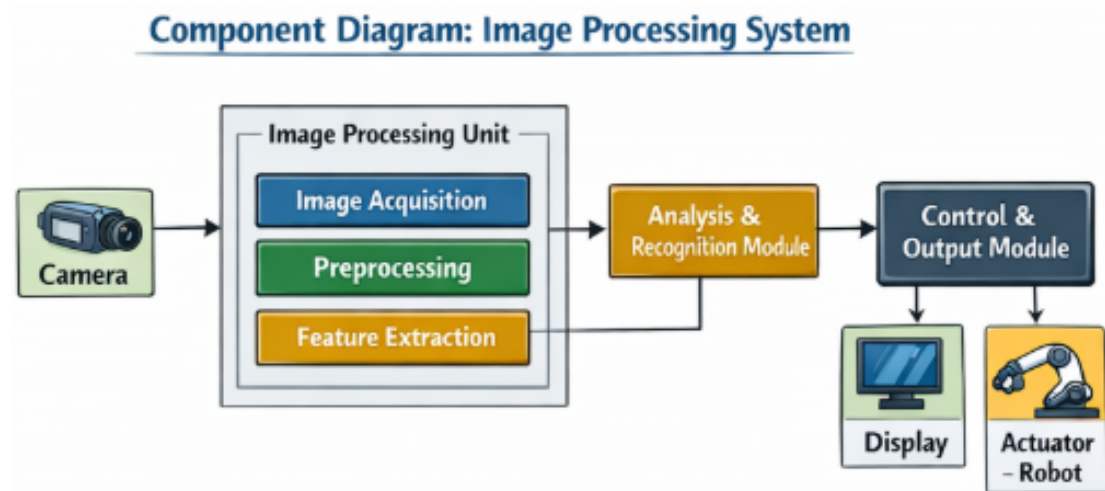


Figure 4.9: Component Diagram

The component diagram shows the physical components of the system and their relationships. It includes modules such as sensor module, data processing module, machine learning module, and alert module. Each component performs a specific function and interacts with others. The diagram highlights how the system is divided into manageable parts. It is useful for system design and implementation planning.

4.3.8 Deployment Diagram

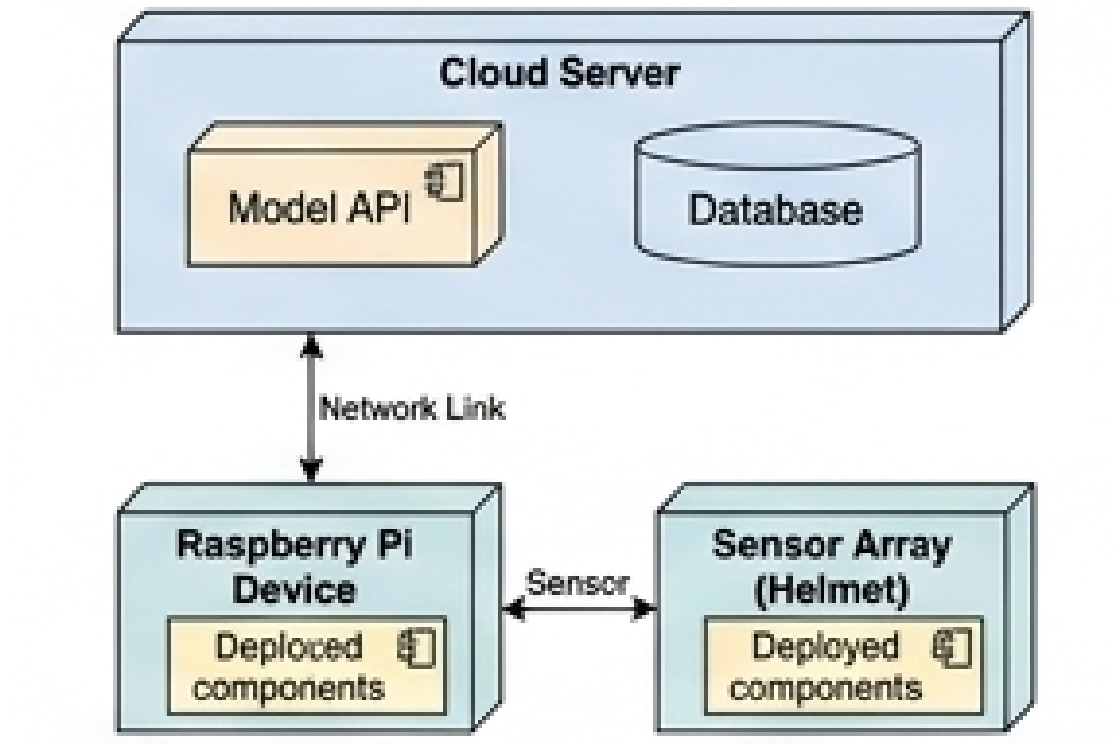


Figure 4.10: Deployment Diagram

The deployment diagram represents the physical deployment of system components on hardware devices. It shows how sensors, processing units, mobile devices, and servers are connected. The diagram illustrates communication between hardware components through networks. It helps in understanding the real-world implementation of the system. This diagram is important for system installation and deployment.

CHAPTER 5

System Implementation

5.1 Modules Description

The proposed bike crash detection system [9016866] follows a modular architecture to ensure efficient, scalable, and reliable performance. It begins with a data acquisition module that collects real-time data from accelerometers and gyroscopes, capturing linear and angular motion. The preprocessing module then cleans the data using techniques like noise filtering, normalization, and segmentation to improve accuracy and consistency.

Next, the feature extraction and classification module uses a hybrid CNN-LSTM model, where CNN extracts spatial features and LSTM captures temporal patterns in the time-series data. This combination enables accurate detection of crash and non-crash events. Finally, the alert generation module activates when a crash is detected, retrieving GPS location and sending emergency notifications to predefined contacts with essential details like location and time.

Overall, this modular design ensures a complete, efficient, and real-time crash detection system with improved safety and quick response capability.

5.2 Technology Used

The proposed bike crash detection system utilizes a combination of modern hardware and software technologies to achieve accurate and reliable performance. It employs sensors such as accelerometers and gyroscopes to capture real-time motion data, including linear acceleration and angular velocity, which help identify abnormal patterns associated with crash events. A GPS module is also integrated to track the vehicle's location and provide real-time coordinates during emergencies.

On the software side, Python is used as the primary programming language

due to its simplicity and strong support for machine learning libraries. Data preprocessing and handling are performed using NumPy and Pandas, ensuring efficient data management and improved model performance. Signal processing techniques are applied to reduce noise and enhance the quality of sensor data.

The system is developed using deep learning frameworks such as TensorFlow and Keras to implement the hybrid CNN-LSTM model. Development and testing are carried out using platforms like Jupyter Notebook or Google Colab for easy experimentation and visualization. Additionally, communication technologies and APIs are used to send real-time alerts and notifications to emergency contacts. Overall, the integration of these technologies ensures efficient, accurate, and real-time crash detection.

5.3 Algorithms used

The proposed system utilizes advanced machine learning algorithms, particularly deep learning techniques, to achieve accurate crash detection. The main algorithm used in this system is a hybrid CNN-LSTM model, which combines the strengths of both Convolutional Neural Networks (CNN) and Long Short-Term Memory (LSTM) networks.

The CNN algorithm is primarily used for feature extraction. It processes multi-dimensional sensor data and identifies spatial patterns such as sudden spikes in acceleration or unusual motion behavior. CNN automatically learns important features without the need for manual feature engineering.

The LSTM algorithm is used for analyzing time-series data. Since crash events occur over a sequence of time, LSTM helps in capturing temporal dependencies and understanding how motion evolves before, during, and after a crash. This makes it highly effective for distinguishing between actual crashes and normal riding conditions.

The overall algorithm follows a structured workflow. First, sensor data is collected and preprocessed. Then, the processed data is fed into the CNN layers for feature extraction. The extracted features are passed to the LSTM layers for sequence analysis. Finally, a classification layer determines whether the event is a crash or not.

If a crash is detected, the system triggers the alert mechanism. The model is trained using labeled data and evaluated using performance metrics such as accuracy, precision, recall, and F1-score. This ensures that the system performs reliably under different conditions.

By combining CNN and LSTM, the system achieves higher accuracy, better generalization, and reduced false positives compared to traditional methods.

5.4 Coding

```
import argparse
from pathlib import Path

import cv2

VALID_EXTENSIONS = {".mp4", ".avi", ".mov", ".mkv"} donot change structure but give

def extract_frames(video_path: Path, class_output_dir: Path, sample_every_n_frame
    capture = cv2.VideoCapture(str(video_path))
    if not capture.isOpened():
        print(f"[WARN] Could not open video: {video_path}")
        return 0

    frame_idx = 0
    saved_count = 0

    while True:
        ret, frame = capture.read()
        if not ret:
            break

        if frame_idx % sample_every_n_frames == 0:
```

```

        out_name = f"{video_path.stem}_f{frame_idx:06d}.jpg"
        out_path = class_output_dir / out_name
        cv2.imwrite(str(out_path), frame)
        saved_count += 1

    frame_idx += 1

capture.release()
return saved_count

def build_dataset(input_dir: Path, output_dir: Path, sample_every_n_frames: int) -
    if not input_dir.exists():
        raise FileNotFoundError(f"Input directory not found: {input_dir}")

    classes = [p for p in input_dir.iterdir() if p.is_dir()]
    if not classes:
        raise ValueError("No class directories found in input directory.")

    output_dir.mkdir(parents=True, exist_ok=True)

    total_saved = 0
    for class_dir in classes:
        class_name = class_dir.name
        class_output_dir = output_dir / class_name
        class_output_dir.mkdir(parents=True, exist_ok=True)

        video_files = [
            p for p in class_dir.rglob("*")
            if p.is_file() and p.suffix.lower() in VALID_EXTENSIONS
        ]

```

```

print(f"[INFO] Class '{class_name}' videos: {len(video_files)}")

    for video_file in video_files:
        saved = extract_frames(video_file, class_output_dir, sample_every_n_frames)
        total_saved += saved

print(f"[INFO] Class '{class_name}' extracted frames: {len(list(class_output_dir.glob('*')))}")

print(f"[DONE] Total extracted frames: {total_saved}")

def parse_args() -> argparse.Namespace:
    parser = argparse.ArgumentParser(description="Extract frames from class-wise videos")
    parser.add_argument("--input-dir", type=Path, required=True, help="Path to raw videos")
    parser.add_argument("--output-dir", type=Path, required=True, help="Path to processed frames")
    parser.add_argument("--sample-every-n-frames", type=int, default=4, help="Frame sampling interval")
    return parser.parse_args()

if __name__ == "__main__":
    args = parse_args()
    build_dataset(args.input_dir, args.output_dir, args.sample_every_n_frames)

```

CHAPTER 6

System Testing

6.1 Testing Methodology

Testing is an essential phase in the bike crash detection system to ensure it works accurately, reliably, and efficiently under different conditions. The main goal is to verify each module, evaluate overall performance, and confirm that the system meets all requirements.

The testing process includes both functional and performance testing. Functional testing checks whether each part of the system—such as data acquisition, preprocessing, model prediction, and alert generation—works correctly. Performance testing evaluates speed, accuracy, and response time when handling real-time data.

The system is tested using both simulated crash scenarios and real-time sensor data from accelerometers and gyroscopes. Simulated data helps create controlled conditions, while real-time data reflects actual riding situations, providing a complete evaluation.

Model performance is measured using key metrics such as accuracy, precision, recall, and F1-score. These metrics help assess how well the system detects crashes and avoids false alarms.

The alert generation module is also tested to ensure it sends instant notifications with accurate GPS location details when a crash occurs, enabling quick emergency response.

Overall, this testing approach ensures that the system is robust, reliable, and effective in real-world conditions.

6.2 Testing under Various Stages

The testing process is carried out in multiple stages to ensure that each component of the system functions correctly and integrates smoothly with

others.

1. Unit Testing Each module is tested individually, including data acquisition, preprocessing, feature extraction, and model prediction, to ensure correct functionality and early error detection.

2. Integration Testing All modules are combined to verify proper interaction and smooth data flow between components, ensuring correct input-output relationships.

3. System Testing The complete system is evaluated to confirm that it meets requirements and accurately classifies crash and non-crash events under different scenarios.

4. Real-Time Testing Live sensor data is used to assess system performance in real-world conditions, ensuring instant crash detection without delays.

5. Performance Testing The system is evaluated based on speed, response time, and accuracy to ensure efficient handling of continuous real-time data.

6. Validation Testing Ensures that the system meets user expectations and performs reliably under various real-world conditions.

6.3 Test Cases

S.No	Test Case	Expected Output	Result
1	Sudden braking	No crash detected	Passed
2	Riding over pothole	No crash detected	Passed
3	Normal riding condition	No crash detected	Passed
4	Simulated crash event	Crash detected	Passed

Table 6.1: Test Cases for Crash Detection System

CHAPTER 7

Results and Discussion

7.1 Introduction

This chapter presents a comprehensive discussion of the results obtained from the implementation and evaluation of the proposed bike crash detection system. The primary objective of this chapter is to analyze how effectively the system meets the goals defined in the earlier stages of the project, particularly in terms of accuracy, reliability, and real-time performance.

The significance of this chapter lies in validating the effectiveness of the hybrid machine learning model used in the system. By examining the results, it becomes possible to assess whether the system can reliably distinguish between crash and non-crash events under different riding conditions. This evaluation is essential to determine the practical applicability of the system in real-world scenarios.

The results are derived from both simulated and real-time sensor data collected using accelerometers and gyroscopes. These sensors provide continuous information about the motion and orientation of the vehicle. The collected data is processed and analyzed using the hybrid CNN-LSTM model, which identifies patterns associated with crash events.

This chapter is organized into multiple sections to provide a structured analysis. The overview of results presents the key findings and performance metrics. The result screenshots section explains the system outputs and their interpretation. Finally, the summary consolidates the major observations and highlights the contribution of the system toward improving road safety.

Overall, this chapter plays a critical role in demonstrating the success of the proposed approach and providing insights into system performance under various conditions

7.2 Overview of Results

The performance of the proposed bike crash detection system was evaluated using both simulated datasets and real-time sensor data collected from accelerometers and gyroscopes. The evaluation primarily focused on measuring the system's ability to accurately classify crash and non-crash events while minimizing false detections.

The results demonstrate that the system achieves a high level of accuracy in identifying crash events. The hybrid CNN-LSTM model effectively learns both spatial and temporal features from the sensor data, which significantly improves classification performance when compared to traditional threshold-based approaches. The system adapts to variations in motion patterns and provides more reliable predictions under different riding conditions.

Overall, the results indicate that the proposed bike crash detection system is highly effective, reliable, and suitable for real-world deployment.

7.3 Result Screenshots

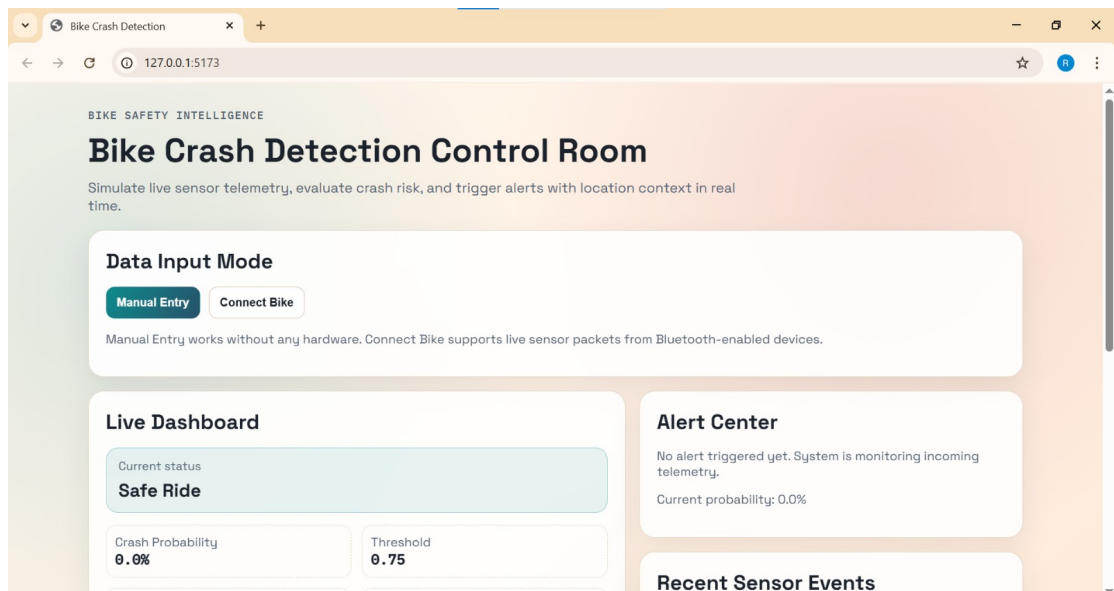


Figure 7.1: Output1

Bike Crash Detection

127.0.0.1:5173

0.0%

0.75

Risk Band
No reading yet

Alert
Standby

Recent Sensor Events
No recent data yet.

Sensor Simulation

Normal Ride **Hard Brake** **Likely Crash**

Accelerometer X
9.5

Accelerometer Y
0.2

Accelerometer Z
9.8

Speed (km/h)
25

Latitude
12.9352

Longitude
77.6245

Upload & Predict

Figure 7.2: Output2

CHAPTER 8

Conclusions and Future Scope

8.1 Conclusions

This project focuses on the development of a machine learning-based bike crash detection system aimed at improving road safety and enabling faster emergency response. Road accidents involving bicycles and two-wheelers are a major concern worldwide, often resulting in serious injuries and fatalities due to delayed medical assistance. Therefore, the need for an intelligent and automated crash detection system is essential. The primary objective of this project was to design and implement a system capable of accurately detecting crash events using sensor data and advanced machine learning algorithms. This objective has been successfully achieved through systematic design, implementation, and evaluation of the proposed model.

The proposed system utilizes data collected from embedded sensors such as accelerometers and gyroscopes, which are commonly available in modern smartphones and wearable devices. These sensors continuously monitor the motion, orientation, and acceleration of the vehicle in real time. The raw sensor data is first subjected to preprocessing steps to remove noise, normalize values, and extract meaningful features. This ensures that the data is suitable for further analysis and improves the overall performance of the model.

To effectively analyze the complex patterns present in the sensor data, a hybrid deep learning approach combining Convolutional Neural Networks (CNN) and Long Short-Term Memory (LSTM) networks is employed. The CNN component is responsible for extracting spatial features from the input data, identifying important patterns and correlations within the sensor readings. On the other hand, the LSTM component captures temporal dependencies by analyzing sequential data over time, making it particularly effective for time-series data such as motion signals. The integration of CNN and LSTM allows

the model to leverage the strengths of both architectures, resulting in improved accuracy and robustness.

The system is designed to distinguish between crash and non-crash events under various real-world conditions, including different speeds, road types, and environmental factors. By learning from labeled training data, the model becomes capable of recognizing crash patterns with high precision. Once a crash is detected, the system can trigger alerts, which can be used to notify emergency services or predefined contacts, thereby reducing response time and potentially saving lives.

In addition to accuracy, the system also emphasizes real-time performance and efficiency, ensuring that crash detection occurs with minimal delay. This makes the system suitable for practical deployment in real-world applications. The use of machine learning enables continuous improvement, as the model can be retrained and updated with new data to enhance its performance over time.

Overall, this project demonstrates the effectiveness of combining sensor technology with deep learning techniques to create an intelligent crash detection system. The successful implementation of the hybrid CNN-LSTM model highlights its capability to accurately identify crash events and contribute to improved road safety and timely emergency response.

8.2 Future Scope of Work

Although the proposed system shows promising results, there is significant scope for further improvement and development. One of the key areas for enhancement is the use of larger and more diverse datasets. Training the model with real-world crash data collected from different environments can improve its accuracy and generalization capability.

Another area for future work is the integration of additional sensors. Including sensors such as magnetometers, pressure sensors, or even cameras can provide more detailed information about vehicle dynamics. By applying sensor fusion techniques, the system can combine data from multiple sources to improve detection accuracy and reliability.

The system can also be enhanced by using more advanced machine learning models. Techniques such as attention mechanisms, transformer models, or other deep learning architectures may further improve the ability to detect complex motion patterns.

From a practical perspective, the system can be integrated into IoT-based devices such as smart helmets or embedded systems in vehicles. Developing a mobile application can also improve usability by allowing users to receive alerts, track location, and manage emergency contacts easily.

Another important direction for future work is the integration of cloud computing. Cloud-based systems can enable real-time data storage, large-scale analysis, and remote monitoring. This would make the system more scalable and suitable for widespread deployment.

The system can also be extended to support other types of vehicles such as cars and bicycles, making it a universal crash detection solution. Additionally, integration with emergency response services can further improve the effectiveness of the system.

This section discusses the potential directions for future research or work that can build upon your project. It explores areas that require further investigation, additional experiments, or improvements. You should suggest topics or questions that remain unanswered, new areas of application, or possible extensions of the current work. This shows the ongoing relevance of the project and provides a roadmap for future research or practical implementation.

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